Week 08 - Tutorial 01

Transcript Language

English

Hello everyone, so in this video let us try to calculate projection of one vector on another

vector. So, let us begin with this example, suppose, I am taking a vector v which is 5

comma 5 comma 5. So, this is a vector on R3. Now, suppose there is another vector u and

I am denoting it by 0 comma 5 comma 5. So, I am taking another vector on R3. And I want

to calculate the projection of vector u on v.

So, projection of vector u in the direction of v this one we have to calculate. So, in

the lecture we have seen the formula to calculate it, so, we have to calculate the inner product

of u and v, inner product of u and v, we also have to calculate the inner product of v and

v. And as we are calculating the projection in the direction of v, so we have to multiply

the vector v with this scalar.

So, as you calculate the inner product of u and v, you will get an scalar if we calculate

inner product v and v, we will get another scalar you have to multiply this scalar with

this vector to get the projection. So, let us calculate. So, what is inner product u

and v? So, inner product u and v which is same as inner product v and u obviously, so

it is given as, so it is nothing but the dot product.

So, we have two vectors here, 0 comma 5 comma 5, and 5 comma 5 comma 5. So, if we calculate

the inner product, if we calculate the dot product, so, we will get 0 into 5 plus 5 into

5 plus 5 into 5. So, what do we get, we will get 0 plus 25 plus 25. So, will get 50. Similarly,

if you calculate inner product of v and v, the same vector, so you will get 25 plus 25

plus 25, which is 75.

So, the scalar you get here is u v by v v. So, this will give you 50 by 75. So, it is

nothing but 2 by 3. Now, in the direction of v, we have to calculate the projection.

So, our projection of u in the direction of v is nothing but 2 by 3 into our v. So, v

is 5 comma 5 comma 5. So, it is 10 by 3, 10 by 3 and 10 by 3.

So, geometrically what we are doing, we are starting with two vectors, suppose, this is

my v and this is my u. So, this is why u vector, this is my v vector and I am taking projection

of u on this vector. So basically, I am calculating this vector with this much magnitude in the

direction of v. So, this is the geometric representation of this. Now, let us try to

visualize this in GeoGebra.

So, this is r 3 we can see here, so at first we will take the vector v which is 5 comma

5 comma 5, so this is the vector v. Now, where is the vector u? So, here is the vector u

which is 0 comma 5 comma 5. Now, we have to take the projection of u on v as we have calculated

it already. So, we can see that this will give us the projection on u in the direction

of v. So, as you can see, this vector along the rejection of v is the projection of u

on the direction of v. So, the dotted line basically gives us a perception how we calculate

the projection.

Now, if we change the this u and v we can see that this projection will also change.

So, now let us try to see this animation where do we change u and v all together and we can

see the projection is changing. So, in this animation we can see how the change of u and

v is changing the projection in the direction of v. So, this animation helps us to visualize

this projection. Thank you.